PROJECT REPORT | SOLAR PV DESIGN & ENERGY YIELD STUDY

11 kWp Grid-Tied Rooftop Solar PV System

Mirpur, Dhaka, Bangladesh (23.80 N, 90.36 E)

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11.0 kWp	15,343 kWh	1,395	83.1 %	1.75 %
DC capacity 20 x 550 Wp	Energy to grid per year	Specific yield kWh/kWp/year	Performance ratio (PR)	Near-shading loss

Summary

This study designs and simulates an 11 kWp grid-tied photovoltaic system for a 20 m x 14 m flat RCC roof in Mirpur, Dhaka. Twenty 550 Wp monocrystalline half-cell modules are installed in three south-facing rows at 23 degrees tilt, connected as two strings of ten to a 9 kW three-phase inverter with two MPPT inputs. The row pitch of 3.49 m was calculated so that no row shades the next between 09:00 and 15:00 on the winter solstice.

A PVsyst 8.1.7 simulation with Meteonorm 9.0 weather data and a 3D model of the roof, water tank and stair room predicts **15,343 kWh per year** delivered to the grid, a specific yield of **1,395 kWh/kWp** and a performance ratio of **83.1%**. An independent hand calculation in Excel gave 15,073 kWh, within 1.8% of the simulation.

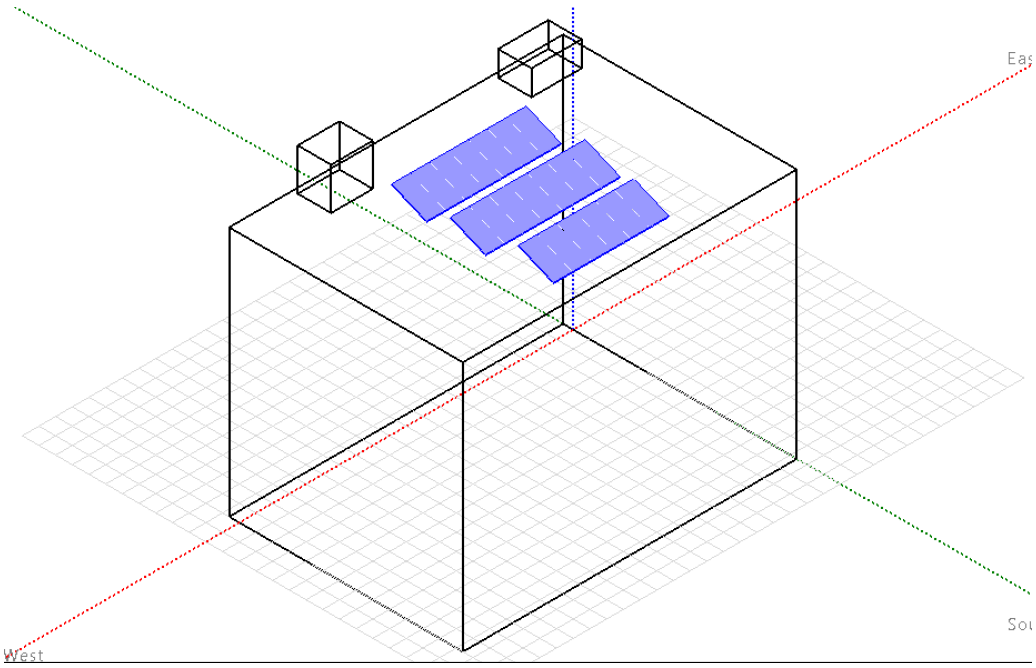


Figure 1. PVsyst 3D near-shading scene: 15 m building, three PV rows, water tank and stair room on the roof.

Tools used: AutoCAD (layout plan and single-line diagram), Microsoft Excel (sizing, cable and energy calculations, BOQ), PVsyst 8.1.7 (3D shading model and hourly energy simulation).

1. Project basis and design workflow

The site is a typical multi-storey residential or commercial building in Mirpur with a flat concrete roof. Two fixed obstructions sit on the north side of the roof: a water tank (3 x 2 x 1.5 m) and a stair room (2.5 x 2 x 2.5 m). A 1 m setback is kept from the parapet on all sides for access and safety.

Parameter	Value	Source
Location	Mirpur, Dhaka, Bangladesh	-
Coordinates / altitude	23.80 N, 90.36 E, 14 m, UTC+6	PVsyst site
Weather data	Meteonorm 9.0, synthetic hourly	PVsyst
Global horizontal irradiation	1,556 kWh/m2 per year (4.26 kWh/m2/day)	Meteonorm
Diffuse horizontal irradiation	863 kWh/m2 per year (55% of global)	Meteonorm
Average ambient temperature	26.3 C (19.0 C Jan to 29.3 C Jun)	Meteonorm
Roof	20 m x 14 m flat RCC, height 15 m	Assumed
Albedo	0.20	PVsyst default

Design workflow

1. **Sizing (Excel):** roof capacity, module count, string voltage and current checks against inverter limits, cable sizing, first energy estimate.

2. **Layout and SLD (AutoCAD):** module placement with setbacks and row pitch, obstruction positions, single-line diagram with protection and earthing.

3. **3D shading model (PVsyst):** building, three PV fields and obstructions; shading animation on 21 December; iso-shading diagram.

4. **Hourly simulation (PVsyst):** energy yield, performance ratio and a full loss breakdown.

5. **Validation:** compare the simulation with the hand calculation and explain the differences.

2. Array layout and row spacing

Twenty modules fit in three rows of 7, 7 and 6, centred on the roof and kept 1 m inside the parapet. Modules are mounted in portrait at 23 degrees (close to the 23.8 degree latitude) facing due south. String A (A1-A10, green) and string B (B1-B10, magenta) each feed one MPPT input.

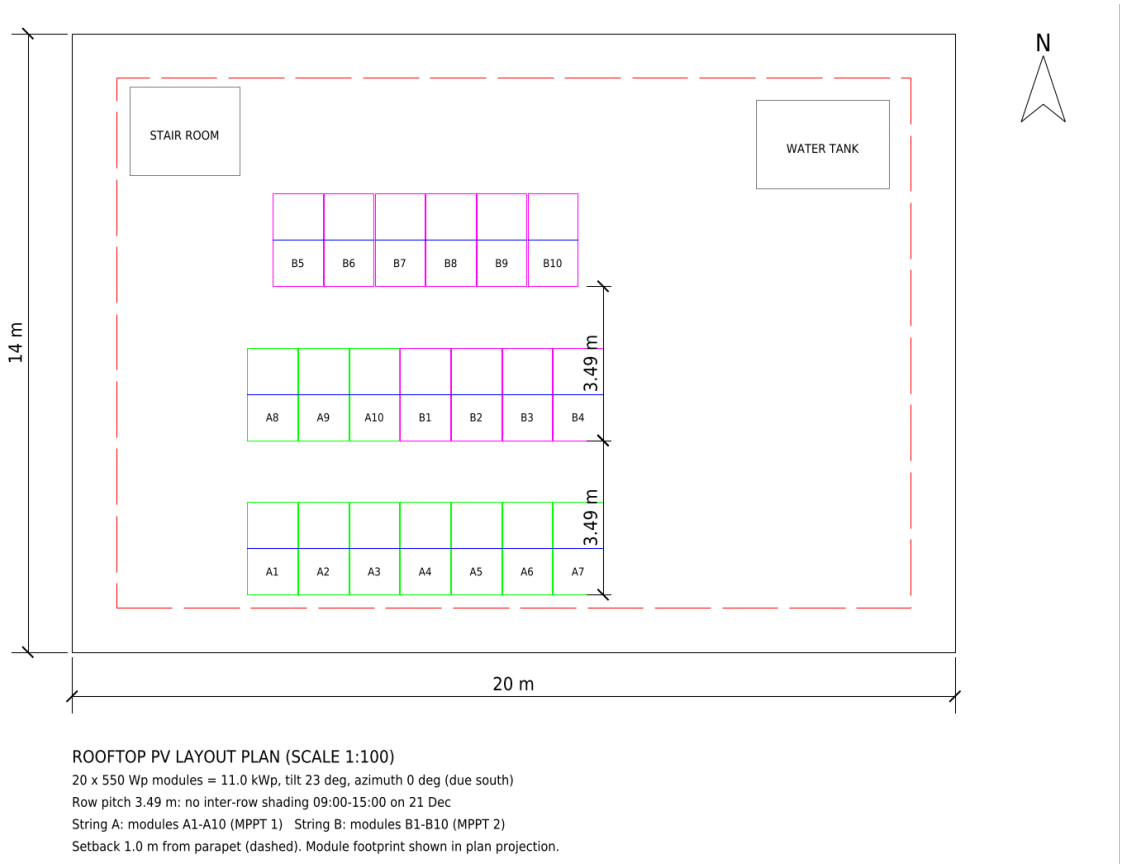


Figure 2. AutoCAD rooftop layout plan (drawing PV-01), plan projection of the tilted modules.

Row pitch calculation

The pitch is set by the worst case: the low winter sun at 09:00 on 21 December, when the sun is at 26 degrees altitude and 42 degrees east of south. A 2.278 m module at 23 degrees has a plan footprint of 2.278 x cos 23 = **2.10 m** and a raised back edge of 2.278 x sin 23 = **0.89 m**. Its shadow along the north-south axis is 0.89 / tan 26 x cos 42 = **1.36 m**. Pitch = 2.10 + 1.36 + small margin = **3.49 m**.

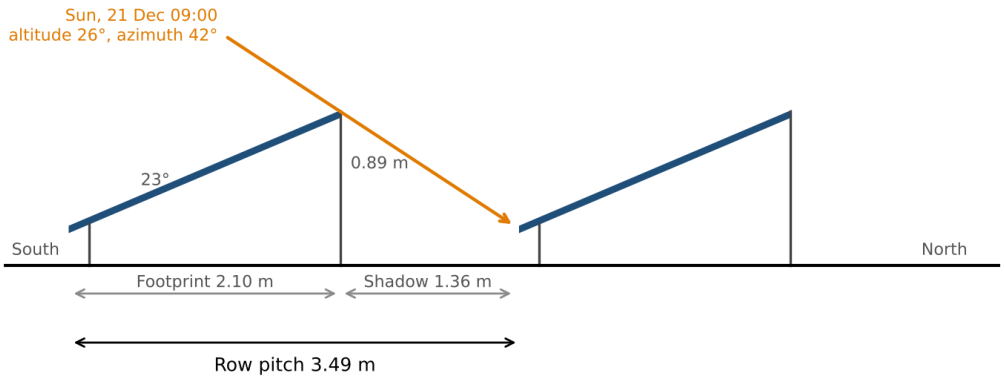


Figure 3. Side elevation of two rows showing how the 3.49 m pitch is derived.

3. Electrical design

Item	Specification
PV module	Mono 550 Wp, twin half-cell (PVsyst generic), 2.28 x 1.13 m, STC efficiency 21.3%
Array	20 modules = 11.0 kWp; 2 strings x 10 in series; module area 51.7 m2
Array at 50 C	Pmpp 10.09 kWp, Umpp 378 V, Impp 27 A
Inverter	9 kWac three-phase, 2 MPPT, operating window 150-750 V
DC/AC ratio	11.0 / 9.0 = 1.22
Grid connection	400 V, 50 Hz, bi-directional net meter (net metering)

String sizing checks

Cold mornings raise the open-circuit voltage; hot afternoons lower the MPP voltage. Both extremes must stay inside the inverter limits.

Check	Calculation	Result	Limit	Status
Max string voltage (8 C)	10 x 49.6 V x [1 + (-0.27%/C)(8 - 25)]	519 V	max 1000 V	OK
Min MPP voltage (cell 70 C)	10 x 41.9 V x [1 + (-0.35%/C)(70 - 25)]	353 V	min 150 V	OK
MPP voltage at STC	10 x 41.9 V	419 V	max 750 V	OK
String design current	1.25 x Isc 14.0 A	17.5 A	Inverter Isc rating	OK
DC/AC ratio	11.0 kWp / 9 kW	1.22	1.0 - 1.3	OK

Cable sizing (copper, voltage-drop limits 1.5% DC and 1% AC)

Circuit	Length	Current	Size	Voltage drop
String A to combiner (DC)	25 m	13.1 A	4 mm2 PV1-F	0.88%
String B to combiner (DC)	30 m	13.1 A	4 mm2 PV1-F	1.06%
Inverter to AC DB (3-ph)	10 m	13.1 A	4C x 6 mm2 Cu	0.21%
AC DB to MDB via net meter	20 m	13.1 A	4C x 10 mm2 Cu	0.26%

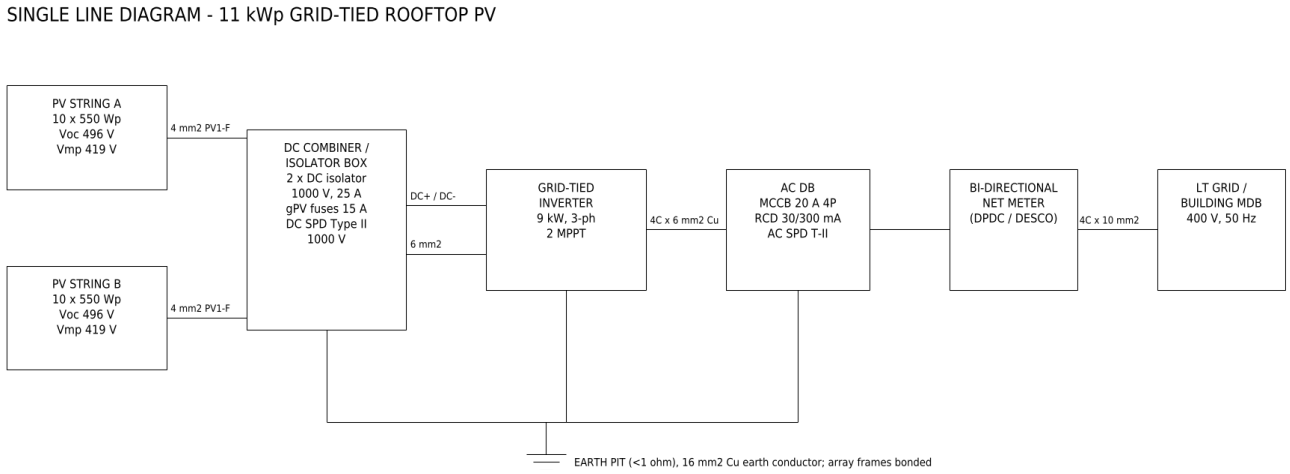


Figure 4. AutoCAD single-line diagram: strings, DC combiner with fuses, isolator and SPD, inverter, AC DB with MCCB, RCD and SPD, net meter and earthing.

4. 3D near-shading analysis

The roof was modelled in PVsyst as a 20 x 14 x 15 m building with three PV fields (7, 7 and 6 modules) raised slightly above the roof surface, plus the water tank and stair room. Positions match the AutoCAD layout. Shading was computed with the linear (irradiance deficit) model.

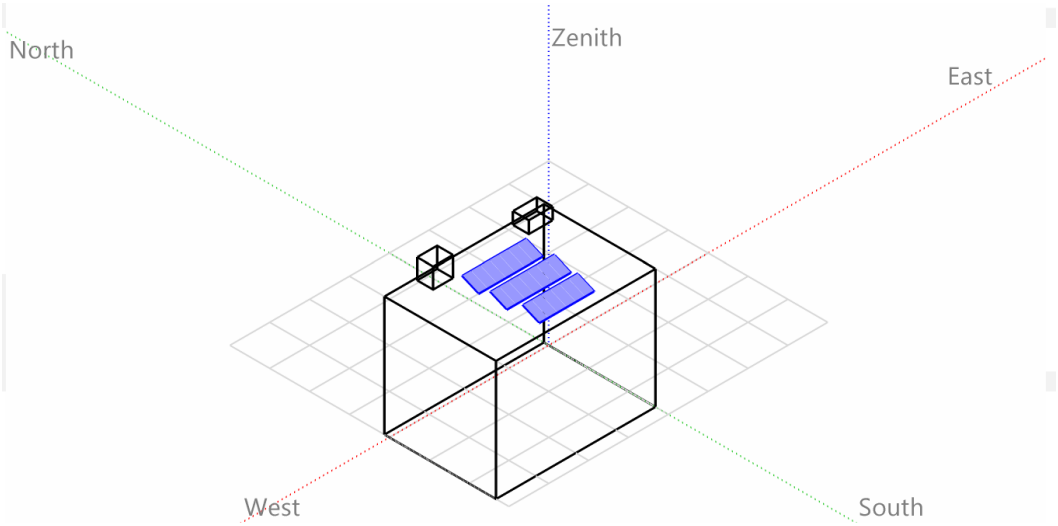


Figure 5. PVsyst perspective of the PV fields and surrounding shading scene.

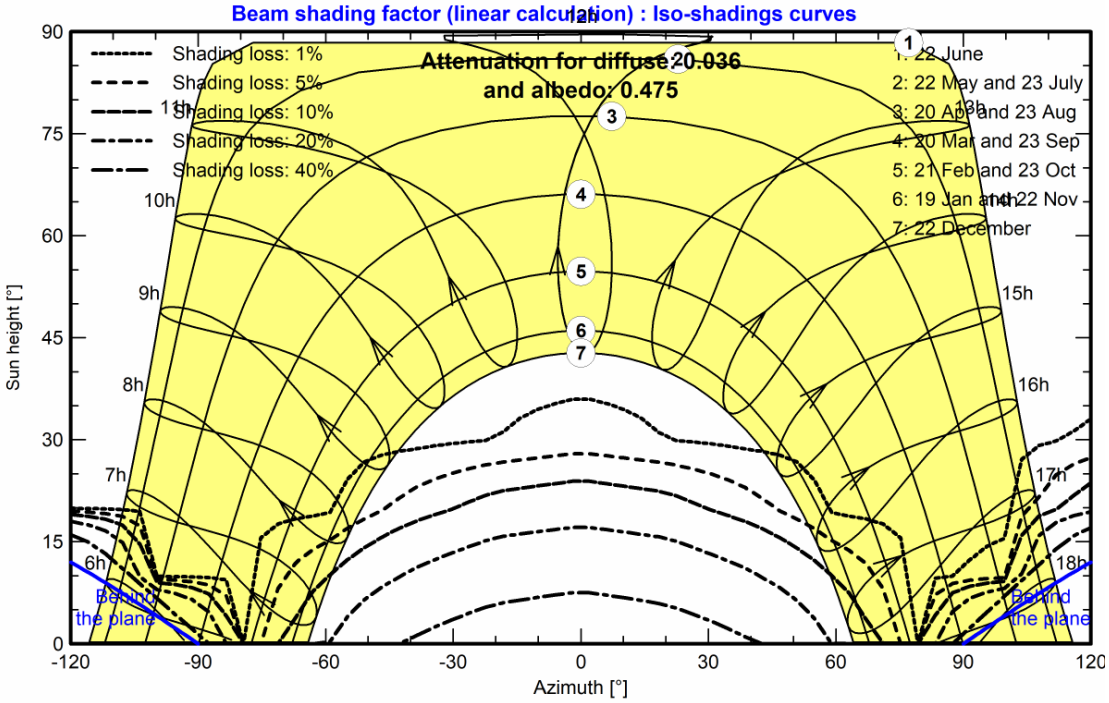


Figure 6. Iso-shading diagram. Shading-loss contours appear only at low sun heights, in the early morning and late afternoon. The midday winter sun path (curve 7, 22 December) is clear.

Result: annual near-shading irradiance loss is only **1.75%**. The diffuse attenuation is 3.6%, mainly from neighbouring rows blocking part of the sky.

Lesson learned: the first model placed the building object above the panels, which PVsyst read as a roof over the array and a 100% shading loss (0 kWh). Checking the side view and the shading factor table caught the error. The building base was reset to ground level with a 15 m height, and the arrays were raised to roof level.

5. Energy simulation results

Result	Value
Energy injected into grid	15,343 kWh/year (about 42 kWh/day on average)
Specific production	1,395 kWh/kWp/year (3.82 kWh/kWp/day)
Performance ratio	83.05% (monthly range 81.4% in June to 85.6% in January)
Global incident on plane	1,679 kWh/m2 per year (+7.9% over horizontal from the 23 degree tilt)
Best / worst month	March 1,672 kWh / July 980 kWh (monsoon cloud cover)

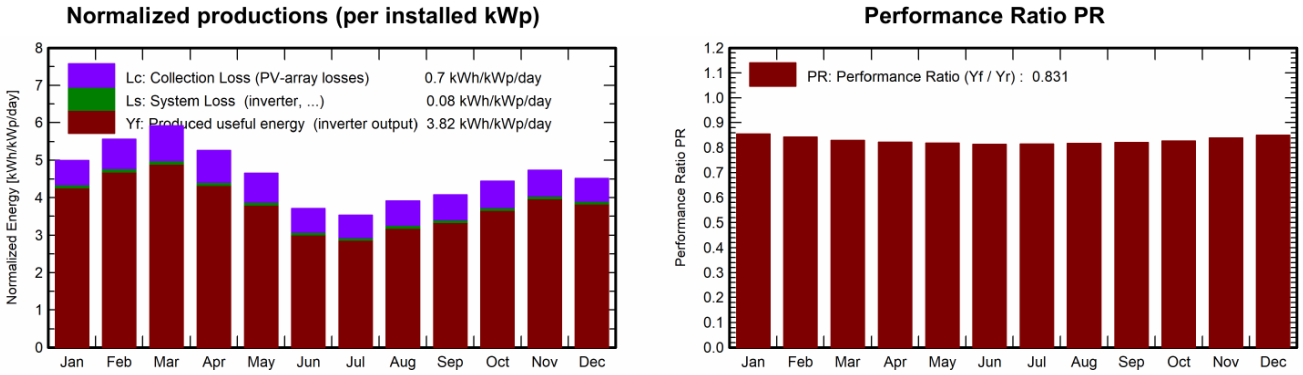


Figure 7. Normalised production per kWp (left) and monthly performance ratio (right), from PVsyst.

Month	GlobHor (kWh/m2)	GlobInc (kWh/m2)	T amb (C)	E_Grid (kWh)	PR
Jan	121.1	154.7	19.0	1,456	0.856
Feb	130.5	155.8	22.7	1,446	0.843
Mar	167.3	183.2	26.9	1,672	0.830
Apr	158.6	158.0	28.7	1,430	0.823
May	152.1	144.2	29.0	1,298	0.818
Jun	120.4	111.2	29.3	996	0.814
Jul	117.3	109.3	29.1	980	0.816
Aug	124.8	121.2	29.3	1,090	0.817
Sep	118.7	122.1	29.0	1,103	0.822
Oct	122.8	137.6	27.8	1,251	0.827
Nov	114.3	142.1	24.4	1,312	0.840
Dec	108.5	140.0	20.6	1,309	0.850
Year	**1,556.3**	**1,679.4**	**26.3**	**15,343**	**0.831**

Source: PVsyst simulation report, balances and main results.

Loss analysis

Starting from 17,251 kWh of array nominal energy at STC efficiency, the largest losses are temperature (6.1%, typical for Dhaka's hot climate), soiling (3.0%, assumed for urban dust), IAM reflection (2.0%), module mismatch (2.0%) and inverter efficiency (1.9%). Near shading costs 1.75%. There is no clipping loss even at a DC/AC ratio of 1.22, so the 9 kW inverter is well matched.

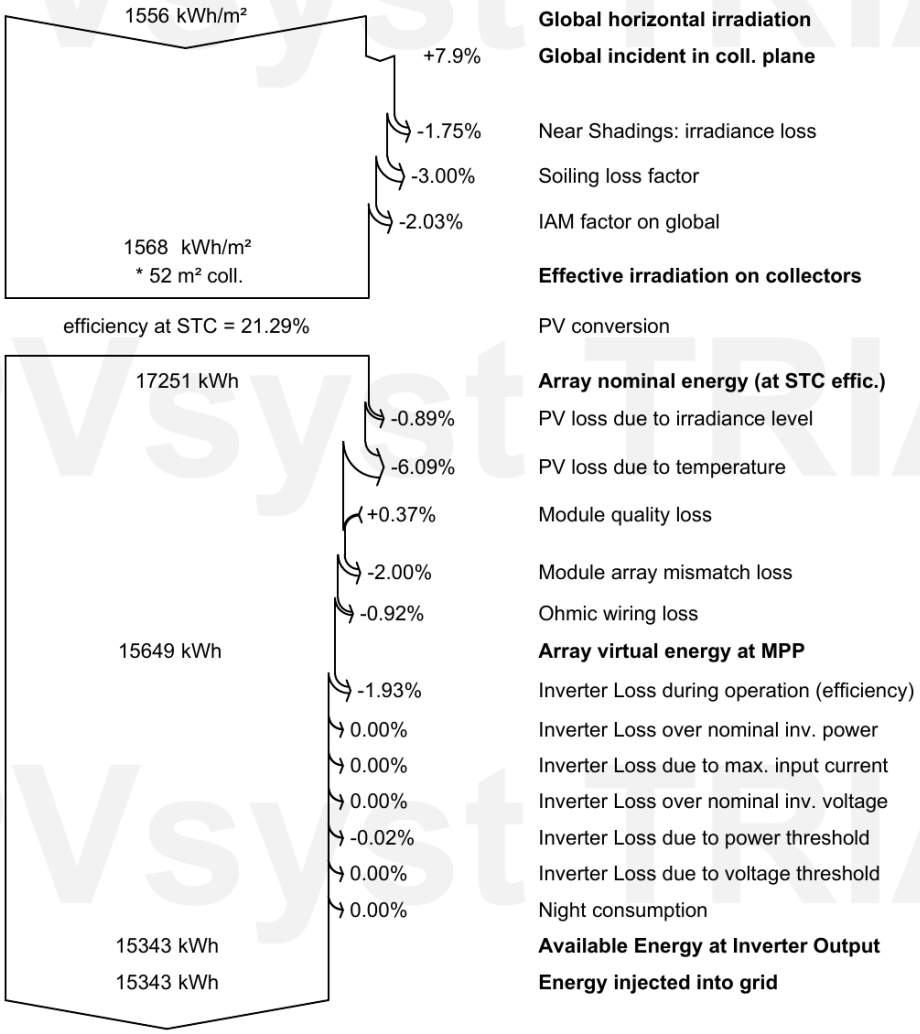


Figure 8. PVsyst loss diagram, from global horizontal irradiation to energy injected into the grid.

6. Validation: hand calculation vs PVsyst

Before simulating, the yield was estimated in Excel as E = capacity x plane-of-array irradiation x PR, using approximate long-term GHI values, a monthly tilt gain factor and an assumed PR of 0.75.

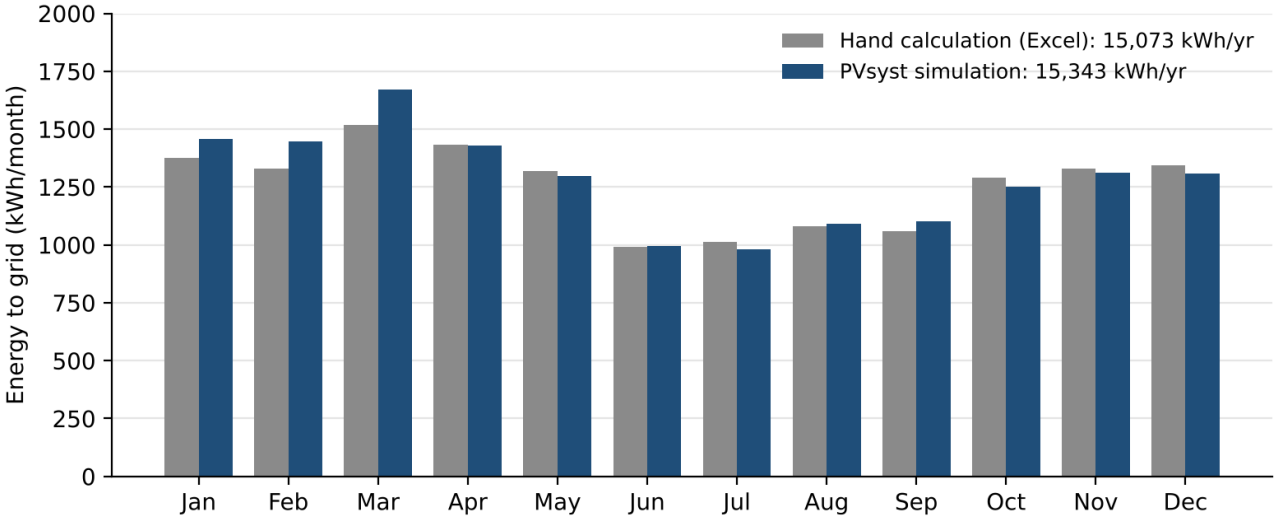


Figure 9. Monthly energy: Excel hand calculation vs PVsyst hourly simulation.

Metric	Hand calculation	PVsyst	Difference
Annual energy	15,073 kWh	15,343 kWh	+1.8%
Specific yield	1,370 kWh/kWp	1,395 kWh/kWp	+1.8%
Performance ratio	0.75 (assumed)	0.831 (computed)	+0.08

Why they differ: the hand estimate used a conservative flat PR of 0.75, while PVsyst computes each loss hour by hour and finds PR 0.831. The two errors partly cancel because the hand estimate also used higher irradiation than Meteonorm in most months. The largest monthly gaps are in January to March, when cooler weather lifts PVsyst's monthly PR to 0.83-0.86, well above the flat 0.75. Agreement within 2% on the annual total confirms the design and the simulation inputs.

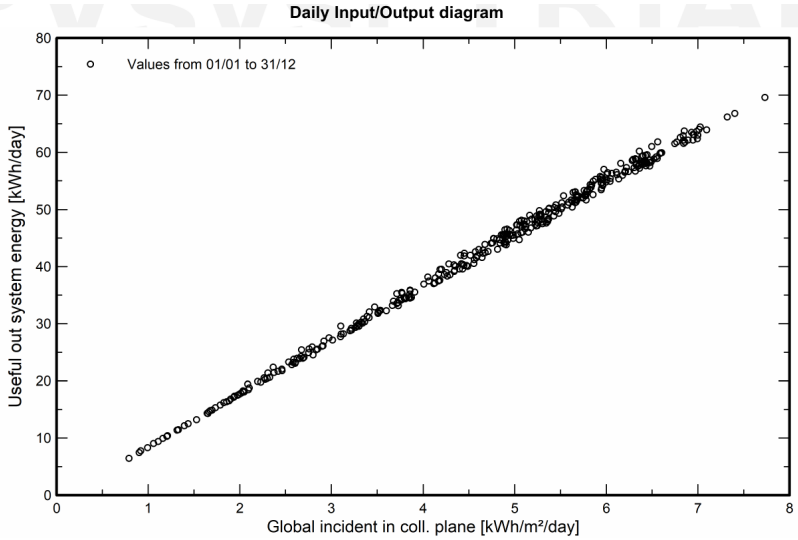


Figure 10. Daily input/output: energy to grid vs daily irradiation on the plane. The straight, tight line shows no clipping or inverter limitation.

7. Bill of quantities

#	Item	Specification	Qty
1	PV module	550 Wp mono half-cell, IEC 61215 / 61730	20 nos
2	Grid-tied inverter	9 kW, 3-phase, 2 MPPT, IEC 62109 / 61727	1 no
3	Mounting structure	Hot-dip galvanised / aluminium, 23 degree fixed tilt	11 kWp
4	DC combiner / isolator box	2-string, gPV fuses 15 A, 1000 V DC isolator, Type II SPD	1 set
5	DC solar cable	4 mm2 PV1-F, red and black (incl. 10% margin)	121 m
6	DC cable	6 mm2 PV1-F, combiner to inverter	10 m
7	MC4 connectors	1000 V, IP67	8 pairs
8	AC cable	4C x 6 mm2 Cu XLPE, inverter to AC DB	11 m
9	AC cable	4C x 10 mm2 Cu XLPE, AC DB to MDB	22 m
10	AC distribution board	MCCB 20 A 4P, RCD, Type II AC SPD, IP65	1 set
11	Earthing	Earth pit, 16 mm2 Cu conductor	2 sets
12	Lightning protection	Air terminal with down conductor	1 set
13	Bi-directional net meter	As per distribution company requirement	1 no
14	Conduits, trays, accessories	UV-resistant PVC conduit, GI tray, glands, lugs	1 lot

Rates are left out; they should come from current vendor quotations.

8. Conclusions and next steps

• An 11 kWp array fits on the 20 x 14 m roof with 1 m setbacks and is expected to deliver about **15.3 MWh per year** (PR 83%), roughly 42 kWh per day on average.

• The 3.49 m row pitch keeps annual near-shading loss to **1.75%**; a tighter pitch would add modules but raise winter losses.

• String voltages stay well inside the inverter window, and the 1.22 DC/AC ratio causes no clipping.

• The hand calculation agreed with PVsyst within 1.8%, which gives confidence in both methods.

• **Next steps:** compare 10 and 15 degree tilt variants for wind load and cleaning; re-run with "according to module strings" shading; add a financial analysis under the 2026 net-metering incentive; and use this hourly PV profile in my ongoing research on PV-integrated EV charging scheduling.

Note: simulation run with PVsyst 8.1.7 in evaluation (trial) mode; module and inverter are PVsyst generic database models. Ratings should be confirmed against the selected manufacturer datasheets before construction.